

# Spatially Dynamic Mixture-of-Experts for Adverse-Weather Salient Object Detection

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**Abstract**— Salient object detection (SOD) degrades under adverse weather. Existing approaches condition on explicit weather labels or apply mixture-of-experts (MoE) to image restoration rather than segmentation. We present SpatialMoE-SOD, routing individual spatial tokens to separate expert networks at three independent pyramid scales without using weather labels during training or inference. A depthwise convolutional router and MLP select top- $K=2$  experts from 8 per scale via true sparse dispatch. Per-token routing entropy is injected as an explicit feature channel into the cross-attention decoder via learnable additive fusion. On WXSOD, the model achieves MAE 0.0192 on 1,500 synthetic test images spanning 9 weather categories and 0.0168 on 554 real-world test images spanning 5 weather categories.

**Index Terms**— salient object detection, mixture-of-experts, adverse weather, sparse routing, cross-attention decoder

## I. INTRODUCTION

SOD identifies visually prominent regions as pixel-wise masks, but performance degrades under adverse weather. Two strategies address this: (1) weather-label-conditioned models—NIFM [1]—require metadata at inference; (2) MoE for weather restoration—WM-MoE [2]—targets pixel regression, not segmentation. To our knowledge, existing MoE-SOD methods primarily explore modality-, scale-, or task-level routing for multi-modal inputs rather than per-token routing for single-RGB SOD. To our knowledge, no existing method combines (a) spatial token-level routing, (b) independent routing at multiple pyramid scales, and (c) an SOD decoder informed by routing confidence, all without weather-specific supervision. Our contributions:

- 1) A token-level spatial router that assigns each token to top- $K=2$  experts from 8 per scale, using depthwise-convolutional context, noisy top- $K$  gating, and true sparse dispatch.
- 2) Three independent routers at scales  $1/4, 1/8, 1/16$ , each with a separate expert pool.
- 3) Routing entropy over the full expert distribution, normalized by  $\log 8$  for  $E=8$ , is injected as an explicit decoder feature via learnable additive fusion.

The rest of the paper is organised as follows. Section II positions the work against adverse-weather SOD and mixture-of-experts methods. Section III describes the architecture: the multi-scale backbone, the spatial router, sparse dispatch, the entropy-fused decoder and the training objective. Section IV reports the results — the headline numbers, a comparison against published methods, and a controlled ablation suite — and explains why the ablation behaves as it does. Section V states what the evidence supports and what it does not.

## II. RELATED WORK

*A Adverse-Weather SOD.* NIFM [1] fuses noise indicators from explicit weather-type one-hot vectors. WXSOD [3] provides 14,945 images with per-image weather labels and synthetic/real-world test splits for weather-robust SOD.

*B MoE for Weather Restoration.* WM-MoE [2] uses weather-aware routing for adverse-weather image restoration, whereas our method performs spatial token-level routing for single-RGB SOD

without weather-specific supervision. Existing MoE weather methods target pixel regression, not segmentation.

*C MoE for SOD.* To our knowledge, existing MoE-SOD methods primarily explore modality-, scale-, or task-level routing for multi-modal inputs rather than per-token routing for single-RGB SOD. V-MoE [4] established token-level top- $K$  routing for classification. Routing entropy has been explored for uncertainty estimation in MoE classifiers. We additionally inject it as an explicit decoder feature for dense prediction, although its isolated contribution has not been evaluated by an ablation.

## III. METHOD

Fig. 1 illustrates the full architecture. A shared PVTv2-B4 backbone extracts multi-scale features; three independent SpatialMoE layers process tokens at each scale; a cross-attention decoder fuses the results and produces dense predictions. We detail each component below.

*A Multi-Scale Features.* A PVTv2-B4 backbone [5] pre-trained on ImageNet extracts features at three scales via `out_indices = (0, 1, 2)`. Each scale is projected to  $C=256$  via  $1 \times 1$  convolution:

$$\mathbf{X}_s = \text{Proj}_s(\text{Backbone}(\mathbf{X})_s) \in \mathbb{R}^{C \times H_s \times W_s}, \quad (1)$$

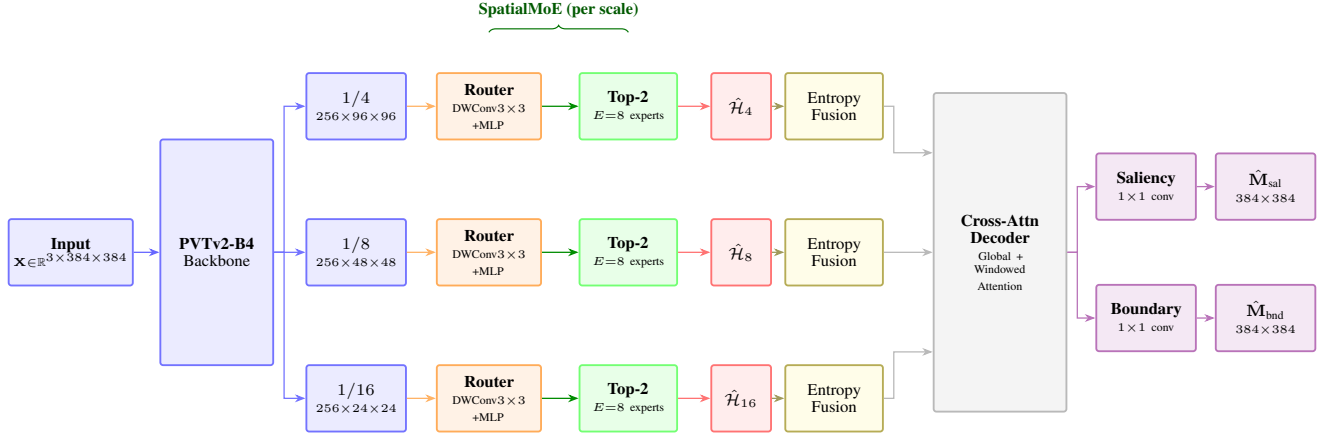
where  $s \in \{4, 8, 16\}$  with  $(H_s, W_s) \in \{(96, 96), (48, 48), (24, 24)\}$ . Each feature map is reshaped into  $N_s = H_s \times W_s$  tokens  $\mathbf{x}_i \in \mathbb{R}^C$ .

*B Router.* For each scale, a depthwise  $3 \times 3$  convolution captures local spatial context from the feature map; the full routing mechanism is shown in Fig. 2. The router input concatenates each token with its local context:

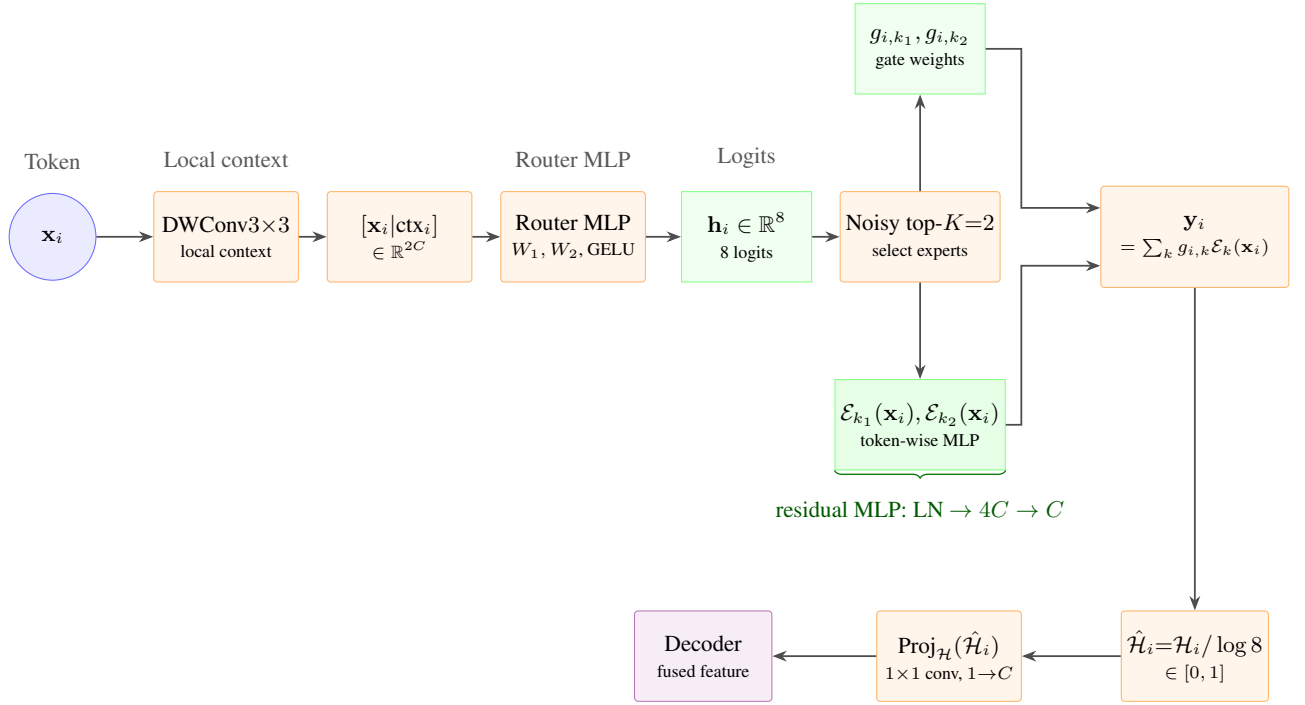
$$\mathbf{r}_i = [\mathbf{x}_i \mid \text{DWConv}_{3 \times 3}(\mathbf{X}_s)_i] \in \mathbb{R}^{2C}, \quad (2)$$

where  $\mid$  denotes channel-wise concatenation, yielding a  $2C$ -dimensional vector that encodes both the token’s content and its spatial neighborhood. A two-layer MLP produces routing logits:

$$\mathbf{h}_i = W_2 \cdot \text{GELU}(W_1 \mathbf{r}_i + b_1) + b_2 \in \mathbb{R}^E, \quad (3)$$



**Figure 1. SpatialMoE-SOD architecture.** Input  $\mathbf{X} \in \mathbb{R}^{3 \times 384 \times 384}$  is processed by a PVTv2-B4 backbone to produce multi-scale features at 1/4, 1/8, 1/16 resolution. Each scale is independently processed by a SpatialMoE layer: a depthwise convolutional router selects top- $K=2$  experts from 8 via noisy gating, and a per-token entropy map  $\hat{\mathcal{H}}$  is computed. The decoder merges scales via global and windowed cross-attention, with entropy features injected via learned additive fusion at each scale. Two  $1 \times 1$  conv heads produce saliency and boundary logits at full resolution.



**Figure 2. Routing mechanism for token  $\mathbf{x}_i$  at one pyramid scale.** A depthwise  $3 \times 3$  convolution extracts local spatial context; the concatenation  $\mathbf{r}_i = [\mathbf{x}_i | \text{ctx}_i] \in \mathbb{R}^{2C}$  is passed through a two-layer MLP to produce 8 routing logits. Noisy top- $K=2$  selection (Gaussian noise during training) picks two experts, branching into gate weights (top) and expert outputs (bottom); gate weights are softmax over selected logits only. The selected experts process the token independently; their outputs are summed with gate weights. Routing entropy  $\mathcal{H}_i \in [0, 1]$  measures assignment uncertainty and is injected as a feature channel into the decoder via a learned  $1 \times 1$  projection.

with  $W_1 \in \mathbb{R}^{128 \times 2C}$ ,  $W_2 \in \mathbb{R}^{E \times 128}$ ,  $E=8$  experts. The hidden dimension 128 provides a bottleneck between the  $2C=512$  input and  $E=8$  output.

*C Noisy Top- $K$  Routing.* During training, Gaussian noise is added with input-dependent scale to encourage exploration:

$$\tilde{h}_{i,j} = h_{i,j} + \epsilon_{i,j}, \quad \epsilon_{i,j} \sim \mathcal{N}(0, \sigma_j^2(\mathbf{x}_i)), \quad (4)$$

where  $\sigma_j(\mathbf{x}_i) = \text{softplus}(w_j^\top \mathbf{x}_i + b_j)$  is a learned per-expert, per-token noise scale. At inference, noise is disabled ( $\epsilon=0$ ). Top- $K=2$  experts are selected per token; gate weights are softmax

over selected logits only:

$$g_{i,k} = \frac{\exp(\tilde{h}_{i,k})}{\sum_{j \in \mathcal{T}_i} \exp(\tilde{h}_{i,j})}, \quad k \in \mathcal{T}_i = \text{top-k}(\tilde{h}_{i,j}, K=2). \quad (5)$$

This produces a sparse assignment: each token attends to exactly 2 of 8 experts, and each expert processes only the tokens that selected it.

*D Sparse Expert Aggregation.* Each expert  $\mathcal{E}_k$  executes only on tokens that selected it. For expert  $k$ , let  $\mathcal{A}_k = \{i : k \in \mathcal{T}_i\}$  denote its assigned tokens:

$$\mathbf{y}_i = \sum_{k \in \mathcal{T}_i} g_{i,k} \cdot \mathcal{E}_k(\mathbf{x}_i), \quad (6)$$

implemented via a loop over experts with mask-based selection (true sparsity, not dense with masking). Each expert is a token-wise residual MLP:

$$\mathcal{E}_k(\mathbf{x}) = \mathbf{x} + W_k^{(2)} \cdot \text{GELU}(W_k^{(1)} \cdot \text{LN}(\mathbf{x})), \quad (7)$$

with  $W_k^{(1)} \in \mathbb{R}^{4C \times C}$ ,  $W_k^{(2)} \in \mathbb{R}^{C \times 4C}$  ( $4\times$  expansion from  $C=256$  to  $4C=1024$ ). Each scale has 8 independent experts with no weight sharing (24 total). Reshaping per-token outputs yields  $\mathbf{Y}_s \in \mathbb{R}^{C \times H_s \times W_s}$ .

*E Routing Entropy.* Per-token routing entropy measures assignment uncertainty over the full expert distribution:

$$\mathcal{H}_i = - \sum_{j=1}^E p_{i,j} \log p_{i,j}, \quad \hat{\mathcal{H}}_i = \mathcal{H}_i / \log 8 \in [0, 1], \quad (8)$$

where  $p_{i,j}$  is the router’s softmax probability for expert  $j$  — computed over all  $E$  experts, not only the selected ones. The normalizing constant is hard-coded to  $\log 8$ , matching the eight-expert model, so  $\hat{\mathcal{H}}_i \in [0, 1]$ : 0 indicates a single expert dominates and 1 a uniform assignment over all eight experts. The normalized entropy map  $\hat{\mathcal{H}}_s \in \mathbb{R}^{1 \times H_s \times W_s}$  is fused with MoE output via a learned additive block:

$$\mathbf{F}_s = \text{Proj}_{\mathcal{H}}(\mathbf{Y}_s) + \lambda_s \cdot \text{Proj}_{\mathcal{H}}(\hat{\mathcal{H}}_s), \quad (9)$$

where  $\text{Proj}_{\mathcal{H}}$  is a  $1 \times 1$  convolution from 1 to  $C$  channels and  $\lambda_s$  is a learnable scalar initialized to 0.1. This allows the decoder to condition its attention on routing confidence without requiring the entropy to be differentiable through the expert weights.

*F Decoder.* The decoder merges coarse-to-fine features via top-down cross-attention. At the coarsest scale,  $\mathbf{F}_{16}$  is upsampled and used as queries with  $\mathbf{F}_8$  as keys and values (8-head global attention):

$$\mathbf{F}'_8 = \text{GlobalCrossAttn}(Q = \text{Up}(\mathbf{F}_{16}), KV = \mathbf{F}_8). \quad (10)$$

Global context is then injected into the fine scale:  $\mathbf{F}_4^{\text{ctx}} = \mathbf{F}_4 + \text{Proj}_g(\text{AvgPool}(\mathbf{F}'_8))$ .  $\mathbf{F}'_8$  is upsampled and used as queries with  $\mathbf{F}_4^{\text{ctx}}$  as keys and values using  $7 \times 7$  windowed attention for efficiency:

$$\mathbf{F}'_4 = \text{WinCrossAttn}(Q = \text{Up}(\mathbf{F}'_8), KV = \mathbf{F}_4^{\text{ctx}}). \quad (11)$$

Three refinement blocks (each:  $\text{Conv } 3 \times 3 \rightarrow \text{GroupNorm} \rightarrow \text{ReLU} \rightarrow \text{Conv } 3 \times 3 \rightarrow \text{GroupNorm} \rightarrow \text{ReLU} \rightarrow \text{residual addition}$ ) progressively upsample from  $1/4$  to full resolution. Two  $1 \times 1$  convolution heads produce saliency logits  $\hat{\mathbf{M}}_{\text{sal}}$  and boundary logits  $\hat{\mathbf{M}}_{\text{bnd}}$  at  $384 \times 384$ .

*G Training Objective.* The objective is a weighted sum of the eight terms that are active in the reported configuration (the Z-loss and the router-confidence term are disabled):

$$\mathcal{L} = \lambda_{\text{BCE}} \mathcal{L}_{\text{BCE}} + \lambda_{\text{IoU}} \mathcal{L}_{\text{IoU}} + \lambda_{\text{SSIM}} \mathcal{L}_{\text{SSIM}} + \lambda_{\text{BND}} \mathcal{L}_{\text{BND}} + \lambda_{\text{LB}} \mathcal{L}_{\text{LB}} + \lambda_{\text{IMP}} \mathcal{L}_{\text{IMP}} + \lambda_{\text{auxBND}} \mathcal{L}_{\text{auxBND}} + \lambda_{\text{DS}} \mathcal{L}_{\text{DS}}, \quad (12)$$

with  $\lambda_{\text{BCE}} = \lambda_{\text{IoU}} = \lambda_{\text{SSIM}} = \lambda_{\text{BND}} = 1.0$ ,  $\lambda_{\text{LB}} = \lambda_{\text{IMP}} = 0.01$ ,  $\lambda_{\text{auxBND}} = 0.5$  and  $\lambda_{\text{DS}} = 0.4$  ( $\lambda_{\text{LB}}$  is applied at each of the three scales).  $\mathcal{L}_{\text{BCE}}$  is binary cross-entropy with logits between

predicted saliency logits and ground-truth mask.  $\mathcal{L}_{\text{IoU}}$  is the soft IoU loss:  $1 - \text{mean}_b \left[ \frac{\sum_{h,w} \sigma(\hat{M}_b) \cdot M_b + \epsilon}{\sum_{h,w} \sigma(\hat{M}_b) + \sum_{h,w} M_b - \sum_{h,w} \sigma(\hat{M}_b) \cdot M_b + \epsilon} \right]$ .  $\mathcal{L}_{\text{SSIM}}$  is 1 minus the structural-similarity index between the sigmoid prediction and the mask.  $\mathcal{L}_{\text{BND}}$  is a smooth- $L_1$  term between the magnitude of the predicted spatial gradient and the ground-truth boundary map.  $\mathcal{L}_{\text{LB}}$  is the load-balancing loss:  $\sum_s E \sum_j f_j^{(s)} P_j^{(s)}$ , summed over the three scales, where  $f_j$  is the fraction of all routing slots assigned to expert  $j$ , computed from hard assignments and detached, and  $P_j$  is the differentiable mean gate mass.  $\mathcal{L}_{\text{IMP}}$  is the importance loss:  $\frac{1}{|S|} \sum_s (\text{std}(\mathbf{P}^{(s)}) / \text{mean}(\mathbf{P}^{(s)}))^2$ , where  $\mathbf{P}^{(s)} \in \mathbb{R}^E$  denotes the per-expert mean gate-mass vector at scale  $s$ .  $\mathcal{L}_{\text{auxBND}}$  is binary cross-entropy between the boundary head’s logits and the boundary map.  $\mathcal{L}_{\text{DS}}$  is the mean binary cross-entropy of the three per-scale auxiliary saliency heads; the auxiliary heads exist because deep supervision is enabled in the reported configuration.

## IV. EXPERIMENTS

*A Setup.* We evaluate on WXSOD [3], which provides 12,891 training images (scene-aware 80/20 split), 1,500 synthetic test images across 9 weather categories, and 554 real-world test images across 5 categories (fog, rain, snow, dark, low-light). Images are aspect-preserving resized to  $384 \times 384$ ; augmentations include horizontal flip ( $p=0.5$ ), color jitter ( $p=0.5$ ), and ImageNet normalization. Training uses DDP on 2 GPUs, AMP (FP16), AdamW optimizer (lr  $10^{-4}$  for both newly introduced modules and the backbone, weight decay  $10^{-4}$ ), WarmupCosine schedule (1% warmup, 50 epochs), gradient clipping at 1.0, effective batch size 32. The backbone is frozen during the first epoch and unfrozen from epoch 1 onward. Metrics: MAE,  $S_\phi$  [6],  $E_\phi^{\text{adp}}$  [7],  $F_\beta^{\text{max}}$  [8]. The model has 69.21M parameters and 277.9G MACs per  $384 \times 384$  input image.

**What we find.** Four results, each backed by a table below.

- **Competitive without weather labels.** MAE 0.0192 on the 1,500 synthesized test images leads every method in the published benchmark, including the weather-supervised WFANet (0.0229); 0.0168 on the 554 real images is second of nineteen (Table 3).
- **The mixture-of-experts does not change the prediction.** Removing it entirely leaves real-test MAE at 0.0195, identical to the full model, and routing at one pyramid scale instead of three gives the same 0.0195 (Table 5).
- **The router is not making a meaningful choice.** Routing entropy sits at 99.8–100% of its log  $E$  ceiling in every arm, and expert weights are orthogonal while their activations correlate at 0.64 (Tables 10 and 12).
- **Training budget matters more than any design choice.** The same recipe at 14 epochs instead of 8 improves real-test MAE from 0.0195 to 0.0174, against a seed spread of  $\pm 0.0004$  (Tables 5 and 13).

*B Main Results.* Table 1 reports global performance on both test splits. The model achieves MAE of 0.0192 on synthetic data and 0.0168 on real-world data. Real-world MAE (0.0168) is slightly lower than synthetic MAE (0.0192). However, this comparison is confounded by test-set composition: the synthetic test set includes compound weather conditions (rain+fog, rain+snow, snow+fog) that are absent from the real-world test set, which

**Table 1.** Global performance on WXSOD test splits.

| Set    | MAE ↓  | $S_\phi$ ↑ | $F_\beta^{\max}$ ↑ | $E_\phi^{\text{adp}}$ ↑ |
|--------|--------|------------|--------------------|-------------------------|
| Synth. | 0.0192 | 0.9139     | 0.9015             | 0.9591                  |
| Real   | 0.0168 | 0.9151     | 0.8936             | 0.9530                  |

**Table 2.** Extended metrics for the reported model. Real split  $N=554$ , synthetic  $N=1500$ .

| Metric                  | Synthetic | Real   |
|-------------------------|-----------|--------|
| $E_\phi^{\text{mean}}$  | 0.9585    | 0.9551 |
| $E_\phi^{\text{max}}$   | 0.9633    | 0.9608 |
| $F_\beta^{\text{mean}}$ | 0.8886    | 0.8747 |
| $F_\beta^{\text{max}}$  | 0.9016    | 0.8936 |
| Boundary MAE            | 0.4817    | 0.6334 |
| Boundary F1             | 0.5472    | 0.3601 |

**Table 3.** Comparison with published WXSOD results. Published values are as reported in [3] (Tables II and III); ours are measured here. WFANet uses weather class labels, our model uses none. Lower MAE and higher  $S_\phi$  are better.

| Method                                   | Synthetic     |               | Real          |               |
|------------------------------------------|---------------|---------------|---------------|---------------|
|                                          | MAE↓          | $S_\phi$ ↑    | MAE↓          | $S_\phi$ ↑    |
| MINet [9]                                | 0.0775        | 0.7403        | 0.0549        | 0.7710        |
| ADMNet [10]                              | 0.0770        | 0.7497        | 0.0544        | 0.7839        |
| TRACER [11]                              | 0.0280        | 0.8898        | 0.0227        | 0.8961        |
| SEANet [12]                              | 0.0332        | 0.8694        | 0.0239        | 0.8978        |
| A3Net [13]                               | 0.0327        | 0.8784        | 0.0263        | 0.8909        |
| DCNet [14]                               | 0.0320        | 0.8792        | 0.0215        | 0.9058        |
| MSRMNet [15]                             | 0.0275        | 0.8895        | 0.0198        | 0.9091        |
| SAFINet [16]                             | 0.0384        | 0.8643        | 0.0217        | 0.9050        |
| TCGNet [17]                              | 0.0317        | 0.8832        | 0.0185        | 0.9185        |
| GeleNet [18]                             | 0.0239        | 0.9038        | 0.0192        | 0.9186        |
| GPONet [19]                              | 0.0266        | 0.8962        | 0.0188        | 0.9098        |
| WFANet [3] ( <i>weather-supervised</i> ) | 0.0229        | 0.9051        | <b>0.0159</b> | <b>0.9248</b> |
| <b>Ours</b> ( <i>no weather labels</i> ) | <b>0.0192</b> | <b>0.9140</b> | 0.0168        | 0.9151        |

may increase synthetic-test error independently of any domain-shift effect.

**C Extended Metrics.** Table 2 reports the remaining metrics for the reported model, and Table 6 gives the same six metrics for every arm in the ablation table. All three E-measure and F-measure variants agree with the trends above. The boundary measures are the weakest component: boundary F1 is 0.3601 on real test, so while the saliency masks are accurate, fine boundary detail is not.

**D Comparison With Published Methods.** WXSOD publishes results for 17 SOD methods and its own weather-supervised baseline WFANet on exactly these two test splits, under the same metric family [3]. Table 3 places our model alongside them. On the synthesized split our model attains the lowest MAE in that table, including WFANet, which is trained with explicit weather-class labels. On the real split it is second of nineteen, behind WFANet and ahead of every other method. The published numbers were produced by their authors under their own protocol; ours are measured here, so this is a like-for-like comparison in dataset, splits and metrics rather than a re-implementation, and it is labelled as reported. Our model is also larger than WFANet (69.21M against 50.87M parameters), which we state plainly rather than present as an efficiency claim.

**E Ablations.** Table 5 reports every arm we trained. The middle block shares one recipe — renormalized gate, router noise enabled, effective batch 32, 8 epochs — and each row changes a single field, so those rows are directly comparable to one another.

Three findings stand out. First, **removing the mixture-of-experts entirely leaves performance unchanged**: the no-MoE arm scores 0.0195 on real test against 0.0195 for the full model, with  $S_\phi$  agreeing to 0.0010. Second, **reducing routing from**

**Table 4.** Arm codes used in Tables 5, 6, 6, 12 and 12. **R** is the reported model; **Ref** is the ablation-family baseline; **A1–A10** change one field from Ref; **B1–B2** use an earlier, non-comparable recipe (batch 40).

| Code                                                              | Configuration                    |
|-------------------------------------------------------------------|----------------------------------|
| <i>Reported model</i> (14 epochs, earlier recipe)                 |                                  |
| <b>R</b>                                                          | E8 $k=2$                         |
| <i>Ablation family</i> (one recipe; single field changed per row) |                                  |
| <b>Ref</b>                                                        | E8 $k=2$ (reference, 8 epochs)   |
| <b>A1</b>                                                         | E8 $k=2$ , 14 epochs             |
| <b>A2</b>                                                         | E2 $k=2$                         |
| <b>A3</b>                                                         | E4 $k=1$                         |
| <b>A4</b>                                                         | E4 $k=2$ , load-balance weight 0 |
| <b>A5</b>                                                         | E4 $k=2$ , dense expert type     |
| <b>A6</b>                                                         | E4 $k=2$ , no MoE at all         |
| <b>A7</b>                                                         | E8 $k=2$ , dense gate            |
| <b>A8</b>                                                         | E8 $k=2$ , routing at 1/4 only   |
| <b>A9</b>                                                         | E8 $k=2$ , seed 43               |
| <b>A10</b>                                                        | E8 $k=2$ , seed 44               |
| <i>Earlier recipe generation</i> (batch 40)                       |                                  |
| <b>B1</b>                                                         | E2 $k=1$ , dense gate            |
| <b>B2</b>                                                         | E4 $k=2$ , dense gate            |

**Table 5.** All trained arms (codes per Table 4). Real split  $N=554$ , synthetic  $N=1500$ .

| Arm        | Real          |               | Synthetic     |               |
|------------|---------------|---------------|---------------|---------------|
|            | MAE↓          | $S_\phi$ ↑    | MAE↓          | $S_\phi$ ↑    |
| <b>R</b>   | 0.0168        | 0.9151        | 0.0192        | 0.9140        |
| <b>Ref</b> | 0.0195        | 0.9076        | 0.0214        | 0.9085        |
| <b>A1</b>  | <b>0.0174</b> | <b>0.9169</b> | <b>0.0202</b> | <b>0.9141</b> |
| <b>A2</b>  | 0.0193        | 0.9087        | 0.0210        | 0.9088        |
| <b>A3</b>  | 0.0212        | 0.8985        | 0.0231        | 0.9036        |
| <b>A4</b>  | 0.0196        | 0.9067        | 0.0212        | 0.9083        |
| <b>A5</b>  | 0.0191        | 0.9086        | 0.0212        | 0.9080        |
| <b>A6</b>  | 0.0195        | 0.9066        | 0.0210        | 0.9084        |
| <b>A7</b>  | 0.0199        | 0.9060        | 0.0212        | 0.9084        |
| <b>A8</b>  | 0.0195        | 0.9077        | 0.0212        | 0.9078        |
| <b>A9</b>  | 0.0199        | 0.9048        | 0.0210        | 0.9086        |
| <b>A10</b> | 0.0191        | 0.9091        | 0.0213        | 0.9083        |
| <b>B1</b>  | 0.0199        | 0.9034        | 0.0215        | 0.9041        |
| <b>B2</b>  | 0.0197        | 0.9043        | 0.0215        | 0.9040        |

**three pyramid scales to one changes nothing** (0.0195), and replacing the routed experts with a single dense expert per scale changes nothing (0.0191). Third, the only architectural knob that moves the metric is top- $k$ : routing to a single expert costs 0.0017 MAE (0.0212 against 0.0195), roughly four times the seed spread of  $\pm 0.0004$  measured across seeds 42–44.

**Training budget is the only variable that matters more than any architectural choice**: the same recipe and seed trained for 14 epochs instead of 8 improves real-test MAE from 0.0195 to 0.0174, a larger effect than every design-choice difference we measured combined. We therefore report the architecture as characterisation, not as a demonstrated improvement.

Table 4 defines a short code for every trained arm; Table 5 and every subsequent per-arm table (Tables 6–12) refer to arms by this code alone, so the full configuration is stated once rather than repeated verbatim in five different tables.

**F Inference-Time Ablations.** Table 7 perturbs the *trained* reference model at inference only, so it isolates how much each component contributes to the final prediction rather than what training does. Replacing the learned router with **random routing** leaves real-test MAE within 0.0005 of the full model. Disabling entropy fusion changes it by 0.0001, and forcing every token at the 1/4 scale to a single expert changes it by  $-0.0001$ . None of the mechanism’s parts affects the output: the mixture, its routing and its entropy channel are all inert at inference.

**G Why Routing Does Not Matter. Why does none of it matter?** Two measurements answer this, and together they are the

**Table 6.** Extended metrics for every arm. Left: real test ( $N=554$ ). Right: synthetic test ( $N=1500$ ). bMAE and bF1 are boundary MAE and boundary F1. Same arm order as Table 5.

| Arm        | Real              |                  |                   |                  |        |        | Synthetic         |                  |                   |                  |        |        |
|------------|-------------------|------------------|-------------------|------------------|--------|--------|-------------------|------------------|-------------------|------------------|--------|--------|
|            | $E^{\text{mean}}$ | $E^{\text{max}}$ | $F^{\text{mean}}$ | $F^{\text{max}}$ | bMAE   | bF1    | $E^{\text{mean}}$ | $E^{\text{max}}$ | $F^{\text{mean}}$ | $F^{\text{max}}$ | bMAE   | bF1    |
| <b>R</b>   | 0.9551            | 0.9608           | 0.8747            | 0.8936           | 0.6334 | 0.3601 | 0.9585            | 0.9633           | 0.8886            | 0.9016           | 0.4817 | 0.5472 |
| <b>Ref</b> | 0.9467            | 0.9567           | 0.8543            | 0.8854           | 0.7211 | 0.2876 | 0.9516            | 0.9617           | 0.8715            | 0.8925           | 0.5534 | 0.4947 |
| <b>A1</b>  | 0.9554            | 0.9631           | 0.8719            | 0.8963           | 0.6675 | 0.3373 | 0.9536            | 0.9617           | 0.8812            | 0.8975           | 0.5067 | 0.5348 |
| <b>A2</b>  | 0.9491            | 0.9583           | 0.8584            | 0.8860           | 0.7186 | 0.2912 | 0.9519            | 0.9611           | 0.8722            | 0.8922           | 0.5521 | 0.4968 |
| <b>A3</b>  | 0.9376            | 0.9560           | 0.8293            | 0.8801           | 0.7688 | 0.2390 | 0.9440            | 0.9592           | 0.8514            | 0.8878           | 0.5874 | 0.4577 |
| <b>A4</b>  | 0.9471            | 0.9571           | 0.8534            | 0.8860           | 0.7260 | 0.2835 | 0.9511            | 0.9604           | 0.8706            | 0.8917           | 0.5544 | 0.4941 |
| <b>A5</b>  | 0.9492            | 0.9581           | 0.8583            | 0.8868           | 0.7174 | 0.2915 | 0.9512            | 0.9602           | 0.8715            | 0.8902           | 0.5540 | 0.4941 |
| <b>A6</b>  | 0.9479            | 0.9581           | 0.8541            | 0.8841           | 0.7250 | 0.2835 | 0.9521            | 0.9615           | 0.8708            | 0.8909           | 0.5557 | 0.4907 |
| <b>A7</b>  | 0.9460            | 0.9566           | 0.8525            | 0.8854           | 0.7317 | 0.2809 | 0.9508            | 0.9608           | 0.8692            | 0.8917           | 0.5612 | 0.4895 |
| <b>A8</b>  | 0.9495            | 0.9592           | 0.8584            | 0.8869           | 0.7221 | 0.2881 | 0.9503            | 0.9588           | 0.8705            | 0.8889           | 0.5554 | 0.4929 |
| <b>A9</b>  | 0.9452            | 0.9552           | 0.8498            | 0.8822           | 0.7345 | 0.2747 | 0.9514            | 0.9609           | 0.8688            | 0.8920           | 0.5559 | 0.4914 |
| <b>A10</b> | 0.9478            | 0.9583           | 0.8565            | 0.8854           | 0.7157 | 0.2922 | 0.9516            | 0.9604           | 0.8714            | 0.8911           | 0.5525 | 0.4945 |
| <b>B1</b>  | 0.9469            | 0.9546           | 0.8535            | 0.8794           | 0.7345 | 0.2739 | 0.9516            | 0.9588           | 0.8687            | 0.8862           | 0.5676 | 0.4782 |
| <b>B2</b>  | 0.9483            | 0.9559           | 0.8561            | 0.8804           | 0.7330 | 0.2768 | 0.9522            | 0.9592           | 0.8692            | 0.8862           | 0.5685 | 0.4771 |

**Table 7.** Inference-time ablations on the reference arm (real test,  $N=554$ ). Random routing is the mean over 4 seeds.

| Inference variant             | MAE↓   | $S_\phi$ ↑ |
|-------------------------------|--------|------------|
| Full model                    | 0.0197 | 0.9043     |
| Random routing (mean of 4)    | 0.0193 | 0.9060     |
| Entropy fusion disabled       | 0.0197 | 0.9044     |
| Force scale 1/4 to one expert | 0.0196 | 0.9048     |

**Table 8.** Routing diagnostics on the reference arm at the final logged epoch. Usage is the hard assignment fraction per expert (uniform would be 12.5% for  $E=8$ ); dead marks experts below 2%. Similarity is the mean pairwise cosine over expert pairs, on real test.

| Scale | Usage % per expert                          | Dead   | Wt. sim. | Act. sim. |
|-------|---------------------------------------------|--------|----------|-----------|
| 1/4   | 21.1, 6.3, 5.3, 5.6, 1.5, 23.8, 29.7, 6.7   | [4]    | +0.0000  | 0.6424    |
| 1/8   | 1.1, 6.9, 3.1, 4.2, 15.5, 0.8, 40.7, 27.7   | [0, 5] | +0.0003  | 0.2733    |
| 1/16  | 9.2, 5.1, 14.1, 13.1, 4.1, 27.1, 16.3, 11.1 | —      | +0.0010  | 0.8617    |

mechanistic contribution of the paper.

*The router is indifferent.* Routing entropy sits at 99.8–100% of its  $\log E$  ceiling (Table 10), so the distribution over experts is essentially uniform: the router almost never expresses a preference.

*The experts are redundant.* Table 8 compares expert pairs at each scale. Pairwise cosine similarity between expert *weights* is essentially zero, so the experts are not copies of one another. But cosine similarity between their *activations* on real data is 0.6424 at scale 1/4, 0.2733 at 1/8 and 0.8617 at 1/16. Despite different weights, the experts’ outputs are substantially correlated at two of the three scales, so for most tokens which expert they are routed to changes little.

Fig. 3 shows the same insensitivity qualitatively, and Table 12 extends the similarity measurement to every arm. This also explains why the one knob that matters is top- $k$ : with  $k=2$  the output averages two correlated experts, suppressing their individual idiosyncrasies, whereas  $k=1$  takes a single expert’s output and is measurably worse (0.0212 against 0.0195).

*H Weather-Wise Analysis.* Table 9 reports per-weather performance on real-world test, and Fig. 4 plots it. Performance varies by  $2.6\times$  across weather types: snow yields the lowest MAE (0.0094,  $N=90$ ), while low-light yields the highest (0.0243,  $N=93$ ). Fog (0.0148,  $N=126$ ), rain (0.0165,  $N=120$ ), and dark (0.0191,  $N=125$ ) fall between these extremes. On synthetic data, compound weather conditions (rain+fog) yield the worst MAE (0.0237), suggesting that compositing multiple degradation types compounds difficulty.

*I Routing Entropy.* Table 10 reports the mean routing entropy per scale. Entropy is taken over the full softmax over all  $E=8$  experts, so the ceiling is  $\log 8 = 2.0794$  nats; we therefore report both the raw value and its fraction of that maximum. The measured mean sits within **0.2%** of the ceiling on every scale on

**Table 9.** Weather-wise results on real-world test (554 images).

| Weather   | $N$ | MAE ↓  | $S_\phi$ ↑ | $F_\beta^{\text{max}}$ ↑ |
|-----------|-----|--------|------------|--------------------------|
| Snow      | 90  | 0.0094 | 0.9478     | 0.9421                   |
| Fog       | 126 | 0.0148 | 0.9179     | 0.8914                   |
| Rain      | 120 | 0.0165 | 0.9157     | 0.8935                   |
| Dark      | 125 | 0.0191 | 0.9072     | 0.8885                   |
| Low-light | 93  | 0.0243 | 0.8897     | 0.8629                   |

both test splits. In other words the router’s distribution over the eight experts is, on average, very nearly uniform: it expresses almost no preference between them. This is a statement about the router’s decisiveness, not about its accuracy—the routing is close to uniform, so the per-token choice of expert carries little information.

**Table 10.** Mean routing entropy per scale. Entropy is over the full distribution over all  $E=8$  experts, so the maximum is  $\log 8 = 2.0794$  nats.

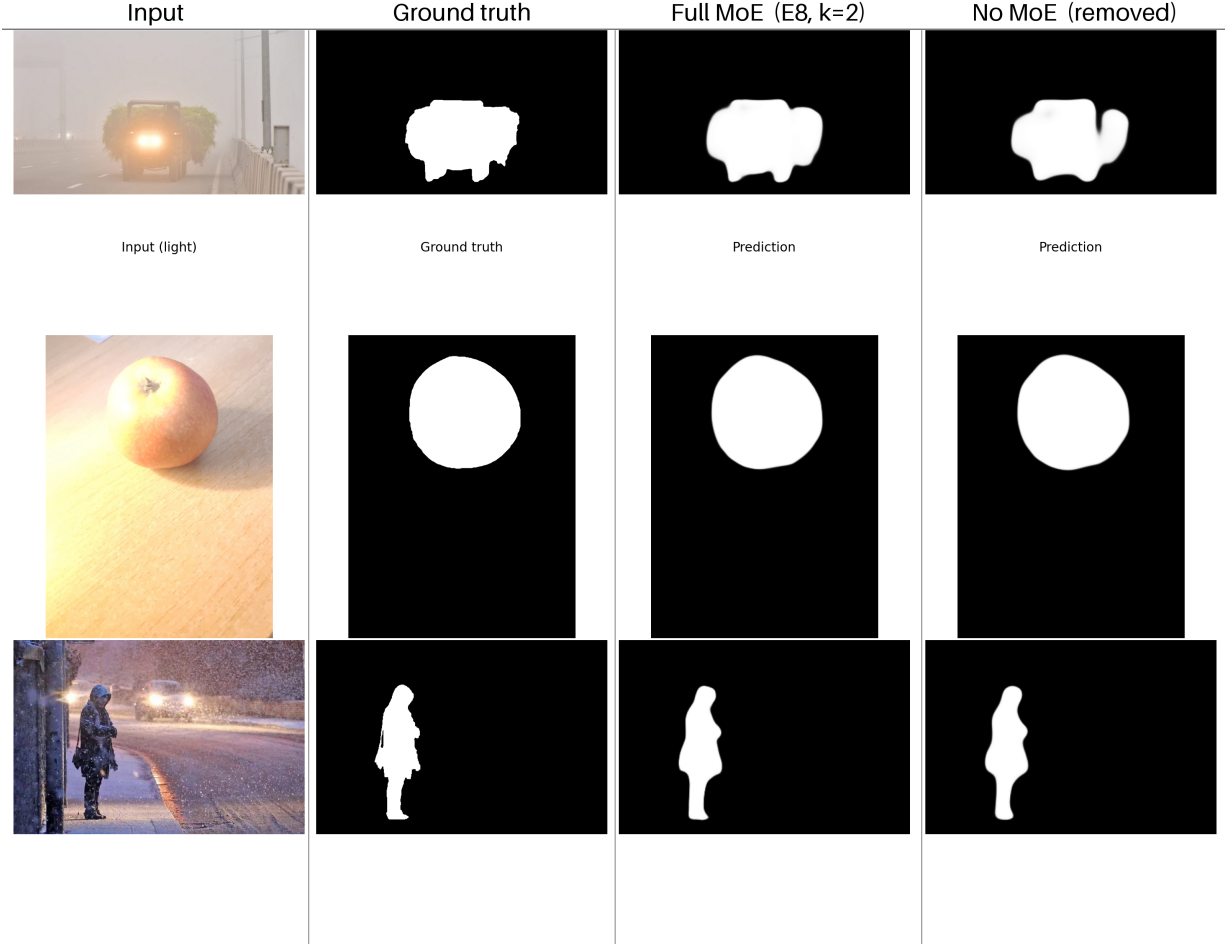
| Scale | Synthetic |          | Real   |          |
|-------|-----------|----------|--------|----------|
|       | nats      | % of max | nats   | % of max |
| 1/4   | 2.0786    | 99.96    | 2.0788 | 99.97    |
| 1/8   | 2.0788    | 99.97    | 2.0789 | 99.97    |
| 1/16  | 2.0755    | 99.81    | 2.0756 | 99.82    |

*J Weather-Wise Analysis: Synthetic.* The synthetic test set spans nine conditions, including three compound ones. Table 11 reports them, ordered worst first. The compound conditions are the hardest, and the clean subset is not the easiest — several single-degradation conditions are harder than the clean images, so the difficulty ordering is not simply a function of how much noise was injected.

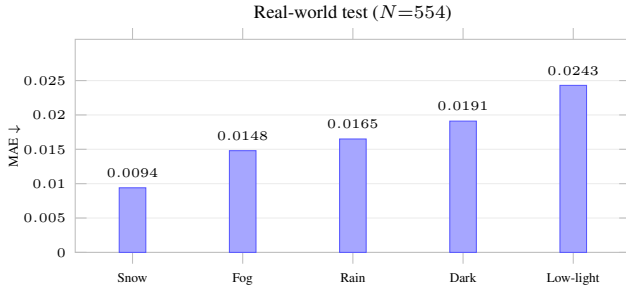
*K Routing Diagnostics for Every Arm.* Table 12 reports, for every trained arm, the normalised routing entropy at each scale and the number of dead experts (usage below 2%). The uniformity is not a property of one configuration: every arm sits at essentially 100% of its  $\log E$  ceiling, including the arms that remove routing entirely at two scales and the arm that removes the load-balancing loss that is supposed to enforce balance.

*L Expert Similarity for Every Arm.* Table 12 reports mean pairwise cosine similarity between expert weights and between their activations. Across every arm, expert *weights* are essentially orthogonal, while expert *activations* are positively correlated — the experts are trained to different parameters that nevertheless respond similarly to real weather images.

*M Seed Variance.* Table 13 repeats one configuration under three seeds. Real-test MAE spans 0.0008 (standard deviation 0.0003) and synthetic MAE spans 0.0004. This is the noise floor for every comparison in this paper: differences smaller than about 0.0005 MAE are not interpretable.



**Figure 3. The mixture-of-experts does not change the prediction.** Read the four columns left to right: (1) input, (2) ground truth, (3) full MoE (E8,  $k=2$ , 14 epochs) and (4) no MoE (the identical architecture with the mixture removed). Columns 3 and 4 are therefore two different models on the same image; both are labelled “Prediction” inside the grid because that is how the evaluation tool renders a single model’s output. Rows are real-world test images under fog, low-light and snow. The two prediction columns are indistinguishable in the low-light row; in the fog and snow rows the differences are confined to object boundaries. This is the qualitative counterpart of the no-MoE row in Table 5, where removing the MoE leaves real-test MAE unchanged at 0.0195.



**Figure 4. Weather-wise MAE on real-world test.** Snow yields the lowest error, while low-light yields the highest.

**Table 11. Weather-wise results on synthetic test (1,500 images), ordered by MAE.**

| Condition | $N$ | MAE↓   | $S_\phi$ ↑ | $E_\phi^{\text{adp}}$ ↑ | $F_\beta^{\text{max}}$ ↑ |
|-----------|-----|--------|------------|-------------------------|--------------------------|
| rainfog   | 172 | 0.0237 | 0.8903     | 0.9384                  | 0.8691                   |
| light     | 166 | 0.0215 | 0.9091     | 0.9623                  | 0.9046                   |
| rain      | 179 | 0.0200 | 0.9173     | 0.9542                  | 0.9030                   |
| dark      | 156 | 0.0199 | 0.9049     | 0.9539                  | 0.8836                   |
| rainasnow | 166 | 0.0194 | 0.9152     | 0.9637                  | 0.9032                   |
| snowfog   | 156 | 0.0189 | 0.9051     | 0.9567                  | 0.8950                   |
| snow      | 172 | 0.0189 | 0.9192     | 0.9647                  | 0.9068                   |
| fog       | 166 | 0.0166 | 0.9308     | 0.9696                  | 0.9255                   |
| clean     | 167 | 0.0142 | 0.9329     | 0.9690                  | 0.9229                   |

**N Discussion.** The model demonstrates that spatial token-level routing is feasible for single-RGB SOD without weather-specific supervision. However, several important limitations must be noted. First, the comparison against prior methods uses the benchmark’s published numbers rather than our own re-implementations, so it is a comparison as reported; WFANet still attains the best real-split MAE, so we do not claim state-of-the-art performance. Second, the model was trained once without replicate runs, so variance estimates are unavailable. Third, the forced-expert analysis suggests that expert specialization may be limited, and the benefit of multi-expert routing over a single larger expert remains unclear. Fourth, routing entropy sits at 99.8–100% of the ceiling over the eight experts, so the decoder’s entropy channel is nearly constant; its contribution has not been isolated by an ablation.

## V. CONCLUSION

We presented SpatialMoE-SOD, which routes individual spatial tokens to separate expert networks at three independent pyramid scales via context-aware routers, without any weather-specific supervision. Per-token routing entropy is normalized and injected as an explicit decoder feature via learnable additive fusion. On WXSOD, the model achieves MAE 0.0192 (synthetic, 1,500 images) and 0.0168 (real-world, 554 images), with

**Table 12.** Diagnostics for every arm. Left: routing entropy per scale, normalised by  $\log E$  (1.000 = a uniform distribution over the eight experts), and the number of dead experts (usage below 2%). Right: mean pairwise cosine similarity between experts, over weight vectors and over activations, real test. R is the reported checkpoint, measured by its own diagnostic pass; the rest are the ablation arms. “—” means not produced for that arm; “n/a” means the quantity is undefined because the scale has fewer than two experts (the pass-through scales of A8, and the single shared expert of A5, A6).

| Arm                 | Routing entropy $H/\log E$ (dead experts) |      |        |      |        |      | Expert similarity |         |         |         |         |          |
|---------------------|-------------------------------------------|------|--------|------|--------|------|-------------------|---------|---------|---------|---------|----------|
|                     | 1/4                                       | dead | 1/8    | dead | 1/16   | dead | wt 1/4            | act 1/4 | wt 1/8  | act 1/8 | wt 1/16 | act 1/16 |
| <b>R (reported)</b> | 0.9997                                    | 0    | 0.9997 | 2    | 0.9982 | 0    | -0.0000           | 0.6383  | +0.0002 | 0.2731  | +0.0011 | 0.8748   |
| <b>Ref</b>          | 0.9996                                    | 1    | 0.9997 | 2    | 0.9980 | 0    | +0.0000           | 0.6424  | +0.0003 | 0.2733  | +0.0010 | 0.8617   |
| <b>A1</b>           | 0.9996                                    | 1    | 0.9998 | 2    | 0.9972 | 0    | +0.0000           | 0.6584  | +0.0003 | 0.2621  | +0.0011 | 0.8830   |
| <b>A2</b>           | 0.9994                                    | 0    | 0.9993 | 0    | 0.9923 | 0    | -0.0002           | 0.7124  | -0.0007 | 0.4914  | +0.0007 | 0.8570   |
| <b>A3</b>           | 0.9983                                    | 1    | 0.9766 | 3    | 0.9933 | 3    | +0.0007           | 0.5687  | -0.0003 | 0.0739  | +0.0011 | 0.8671   |
| <b>A4</b>           | 0.9996                                    | 0    | 0.9998 | 0    | 0.9969 | 0    | +0.0007           | 0.6561  | -0.0002 | 0.2998  | +0.0010 | 0.8556   |
| <b>A5</b>           | 0.0000                                    | 3    | 0.0000 | 3    | 0.0000 | 3    | —                 | —       | —       | —       | —       | —        |
| <b>A6</b>           | 0.0000                                    | 3    | 0.0000 | 3    | 0.0000 | 3    | —                 | —       | —       | —       | —       | —        |
| <b>A7</b>           | 1.0000                                    | 0    | 1.0000 | 0    | 0.9996 | 0    | +0.0000           | 0.7013  | +0.0002 | 0.5607  | +0.0015 | 0.9438   |
| <b>A8</b>           | 0.9996                                    | 1    | 0.0000 | 7    | 0.0000 | 7    | +0.0000           | 0.6424  | —       | —       | —       | —        |
| <b>A9</b>           | 0.9995                                    | 1    | 0.9998 | 0    | 0.9972 | 1    | -0.0005           | 0.6472  | -0.0004 | 0.2631  | +0.0002 | 0.8622   |
| <b>A10</b>          | 0.9995                                    | 1    | 0.9997 | 1    | 0.9973 | 0    | +0.0001           | 0.6412  | -0.0001 | 0.2790  | +0.0006 | 0.8614   |
| <b>B1</b>           | 1.0000                                    | 0    | 1.0000 | 0    | 0.9979 | 0    | -0.0002           | 0.6413  | -0.0008 | 0.0795  | +0.0008 | 0.8908   |
| <b>B2</b>           | 1.0000                                    | 0    | 1.0000 | 0    | 0.9974 | 0    | +0.0006           | 0.6351  | -0.0002 | 0.4469  | +0.0011 | 0.8669   |

**Table 13.** Seed variance for E8  $k=2$ , 8 epochs, everything else fixed.

| Seed | Real   |          |                  | Synthetic |          |                  |
|------|--------|----------|------------------|-----------|----------|------------------|
|      | MAE    | $S_\phi$ | $E^{\text{adp}}$ | MAE       | $S_\phi$ | $E^{\text{adp}}$ |
| 42   | 0.0195 | 0.9076   | 0.9402           | 0.0214    | 0.9085   | 0.9493           |
| 43   | 0.0199 | 0.9048   | 0.9380           | 0.0210    | 0.9086   | 0.9481           |
| 44   | 0.0191 | 0.9091   | 0.9412           | 0.0213    | 0.9083   | 0.9513           |

weather-wise MAE ranging from 0.0094 (snow) to 0.0243 (low-light).

Several limitations warrant future investigation. The model was trained once without replicate runs, so variance estimates are unavailable. Against the benchmark’s published results the model leads the synthesized split and places second of nineteen on the real split, behind the weather-supervised WFA Net [3], while using no weather labels. The forced-expert analysis suggests limited differentiation among experts at scale 1/4, and the contribution of the entropy fusion mechanism relative to a baseline without it has not been ablated. Routing entropy sits at 99.8–100% of the maximum attainable over the eight experts, so routing is close to uniform and the entropy channel carries little signal. The ablation matrix is otherwise complete: removing the mixture-of-experts, reducing routing to one pyramid scale, replacing routed experts with a dense expert and removing the load-balancing loss all leave real-test MAE within 0.0004 of the reference, while routing to a single expert costs 0.0017. The mixture is therefore not earning its parameters on this task, and the useful direction is not more routing capacity but the training budget. Future work should determine whether a routing objective that is actually supervised – rather than one optimised only through the saliency loss – can induce specialisation, and should re-run the benchmark’s baselines under a single unified protocol.

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